De La Salle Holy Cross College

CYCLE TEST

30th September 2024

Grade 10

MATHEMATICS

Functions and Financial Mathematics

Examiner: Mrs Seewald
Moderators: Mrs De Freitas & Mr Guilhermino

Time: 1 hour
Marks: 50

Name:	Memo		
Teacher:	GdF	AS	PG

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 8 pages. Please check that your question paper is complete.
2. Answer all the questions in the space provided on the question paper.
3. A non-graphical, non-programmable calculator may be used.
4. All necessary calculations must be shown clearly. Answers only will not necessarily be awarded full marks.
5. Diagrams and graphs are not necessarily drawn to scale.
6. Round off final answers to TWO decimal places, unless stated otherwise.
7. It is in your own interest to write legibly and to present your work neatly.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	**TOTAL**	
Mark								
Total	9	5	8	12	11	5	**50**	%

Remember that God is with you and will help and inspire you in all that you do

QUESTION 1

Formulae: $A = P(1 + i \times n)$ $A = P(1 + i)^n$

(a) Seven years ago, Kathy decided to invest R 18 000 in a bank account that paid simple interest at 4,5% p.a.

Calculate how much interest Kathy has earned over the seven years. (3)

$A = P(1+i\times n)$

$= 18000\left(1 + \frac{4,5}{100} \times 7\right)$ ✓

$= R23670$ ✓

∴ Earned R 5670 interest ✓a

(b) Kathy wants to buy a television set that costs R 27 660,00 now. If the average rate of inflation over the last three years was 6,7% p.a., calculate the cost of the television set three years ago. (3)

$A = P(1+i)^n$

$27660 = P\left(1 + \frac{6,7}{100}\right)^3$ ✓ sub into correct formula

$\frac{27660}{\left(1 + \frac{6,7}{100}\right)^3} = P$ ✓m

$R22769,78 = P$ ✓a

1/3 not correct formula

(c) At what rate of simple interest should Kathy have invested her money seven years ago if she intends buying the television set now for R27 660 using only her investment of R 18 000 and the interest earned over the last seven years? (3)

$A = P(1+i\times n)$

$27660 = 18000\,(1 + i\times 7)$

$\left(\frac{\frac{27660}{18000} - 1}{7}\right) = \frac{7i}{7}$ ✓m

$0,0766\ldots = i$

$7,66\dot{6}\,\%$

∴ 7,67 % ✓

[9]

QUESTION 2

Given $g(x) = -2x + 3$.

(a) Determine $g(0)$. (1)

$g(0) = -2(0) + 3$ ✓

$g(0) = 3$

(b) Determine x if $g(x) = 0$. (2)

$-2x + 3 = 0$

$\frac{3}{2} = \frac{2x}{2}$ ✓

$\therefore \frac{3}{2} = x$ ✓

(c) Explain, in your own words, what your answers in (a) and (b) mean. (2)

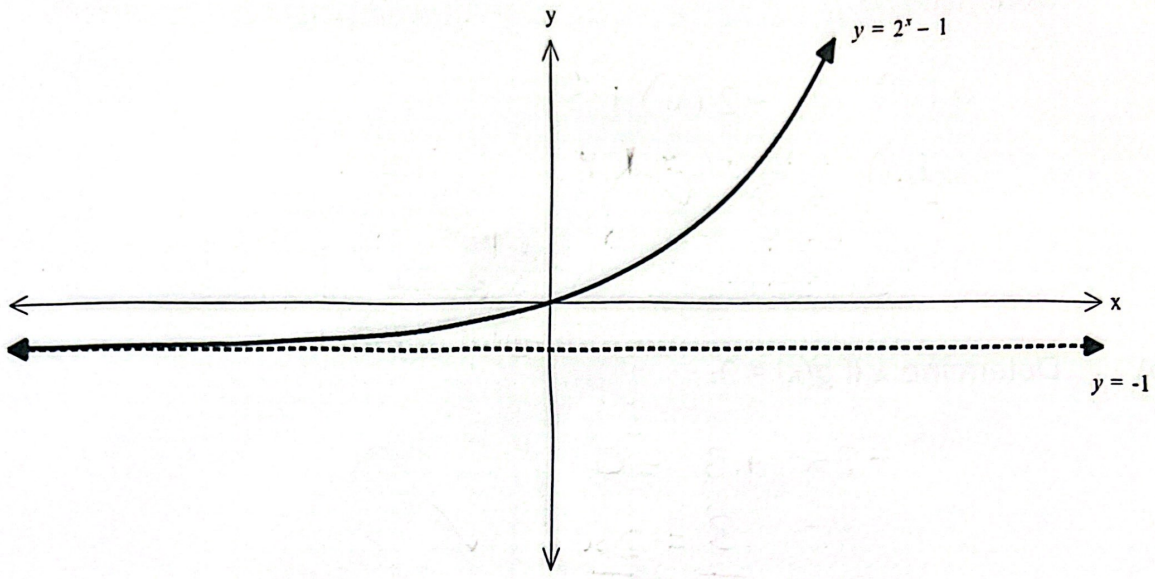
The y-intercept is $(0; 3)$ ✓

The x-intercept is $(\frac{3}{2}; 0)$ ✓

[5]

QUESTION 3

Consider the graph $m(x) = 2^x - 1$ and use it to determine:



(a) the *y*-intercept. (1)

$(0;0)$ ✓

(b) the domain. (1)

$x \in \mathbb{R}$ ✓

(c) the range. (1)

$y > -1$ ✓

(d) Write down the *x*-value(s) for which $m(x) \geq 0$. (1)

$x \geq 0$ ✓

(e) If the graph $m(x) = 2^x - 1$ is shifted down 2 units, determine the equation of the new graph. (2)

$m_{new}(x) = 2^x - 3$ ✓ ✓

same

(f) The graph of $m(x) = 2^x - 1$ is reflected in the x-axis. Write down the new equation of m. (2)

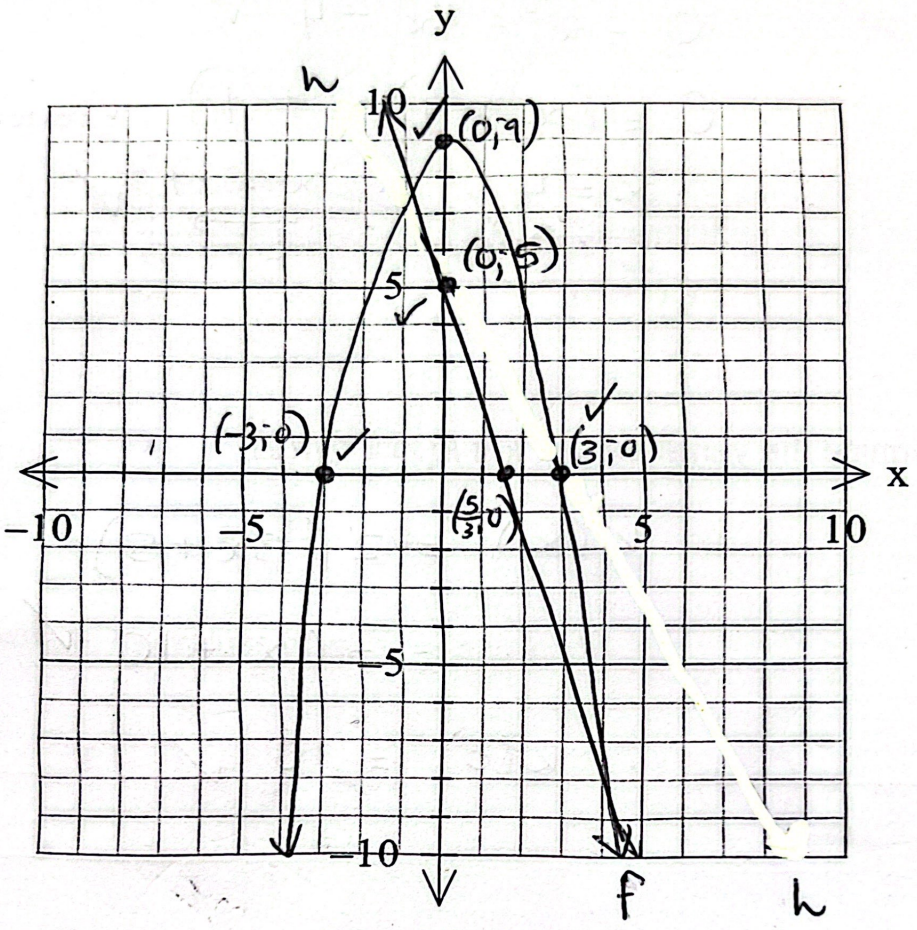
$m_{new}(x) = -2^x + 1$ ✓ ✓

[8]

QUESTION 4

Given functions $f(x) = -x^2 + 9$ and $h(x) = -3x + 5$.

(a) Draw f and h on the same system of axes below. Show all intercepts with the axes. (5)



$f(x) = -x^2 + 9$

x-int: $y = 0$ y-int: $y = 9$

$\therefore 0 = -x^2 + 9$

$x^2 - 9 = 0$

$(x+3)(x-3) = 0$

$\therefore x = -3$ or $x = 3$

$h(x) = -3x + 5$

x-int: $y = 0$

$0 = -3x + 5$

$3x = 5$

b) Solve for x if $f(x) = h(x)$. (5)

$f(x) = h(x)$

$-x^2 + 9 = -3x + 5$ ✓

$0 = x^2 - 3x - 4$ ✓

$0 = (x - 4)(x + 1)$ ✓ factorise

$\therefore x = 4$ ✓ or $x = -1$ ✓

c) Determine the y-intercept of k if $k(x) = 2h(x)$. (2)

$k(x) = 2(-3x + 5)$

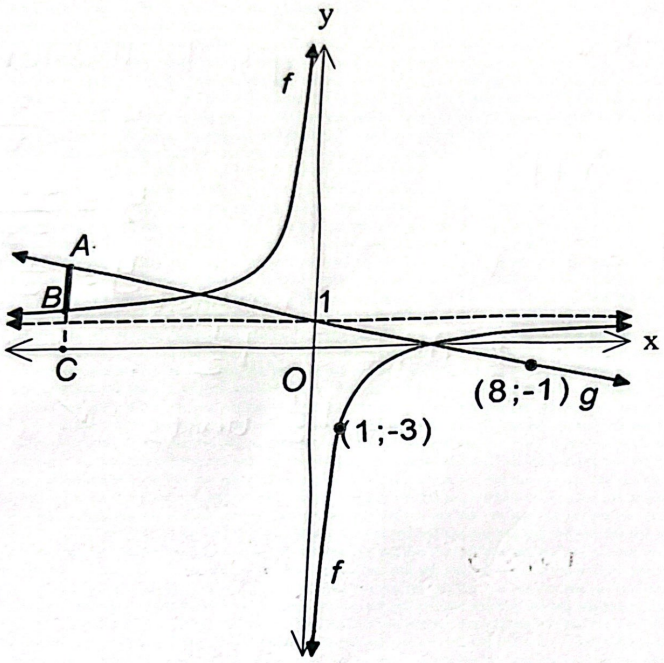
$= -6x + 10$ ✓

$\therefore k(0) = 10$ ✓

[12]

QUESTION 5

In the figure, $f(x) = \frac{a}{x} + q$ and $g(x) = mx + c$.



(a) Determine the values of a, q, m and c. (6)

$f(x) = \frac{a}{x} + 1$

$\therefore q = 1$ ✓

$y = \frac{a}{x} + 1$

Sub $(1;-3)$: $-3 = \frac{a}{1} + 1$ ✓m

$-3 = a + 1$

$-3 - 1 = a$

$-4 = a$ ✓

$f(x) = \frac{-4}{x} + 1$

$g(x) = mx + c$

$c = 1$ ✓

$y = mx + 1$

sub $(8;-1)$: $-1 = m(8) + 1$ ✓m

$-1 = 8m + 1$

$\frac{-2}{8} = \frac{8m}{8}$

$-\frac{1}{4} = m$ ✓

$\therefore g(x) = -\frac{1}{4}x + 1$

(b) Calculate the distance AB if the distance OC = 8 units. (5)

✓$x=-8$

C $(-8;0)$

pt A lies on str. line

$y=-\frac{1}{4}x+1$

$y=-\frac{1}{4}(-8)+1$

$y=2+1=3$ ✓

pt B lies on hyperbola.

$y=\frac{-4}{x}+1$

$y=\frac{-4}{-8}+1$

$y=\frac{1}{2}+1=1\frac{1}{2}$ ✓

✓subtract

$\therefore$ 距 AB $= 3-1\frac{1}{2}$

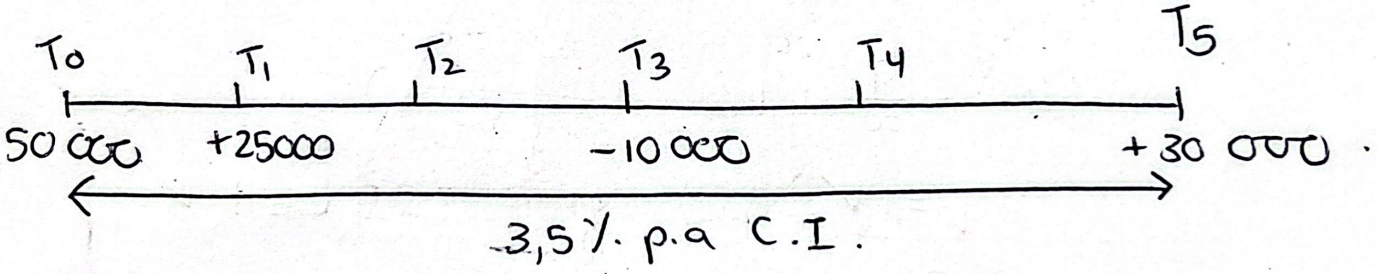
$= 1\frac{1}{2}$ units ✓

[11]

QUESTION 6

Archie opens a savings account with a deposit of R50 000. A year later he deposits R 25 000. Another 2 years later he withdraws R 10 000. Five years after the account was opened, she makes a final deposit of R30 000. The account pays 3,5% p.a. compound interest.

How much money is in the account after the final deposit of R30 000? (5)



$A = P(1+i)^n$ ✓

$= 50\,000(1+3{,}5\%)^5 + 25000(1+3{,}5\%)^4 -$ ✓ ✓

$10\,000(1+3{,}5\%)^2$ ✓ $+ 30\,000$ ✓

$= R107\,360{,}14$ ✓

TOTAL: 50